

Dynamic spatial and temporal prediction of changes in depositional processes on clastic shorelines: Toward improved subsurface uncertainty reduction and management

R. Bruce Ainsworth, Boyan K. Vakarelov, and Rachel A. Nanson

ABSTRACT

Existing classification schemes and models for clastic coastal depositional systems do not consider the potential amplifying or moderating effects of coastal morphology on depositional processes and do not provide a mechanism for the dynamic prediction of changes in coastal depositional style. A new process-based classification scheme based on the relative importance of primary, secondary, and tertiary processes is presented. This scheme permits a semiquantitative classification of clastic coastal depositional systems. In addition, it provides the basis for new models for clastic shorelines that convolve the effects of basin shape, coastal morphology, accommodation space, sediment supply, shoreline trajectory, and shelf width parameters on depositional processes. The end result is a marked improvement in the predictive capabilities of models. The models can describe and predict the likelihood of primary, secondary, and tertiary depositional processes acting in shoreline depositional environments via either a matrix or a decision tree approach. They are also dynamic in nature and can be applied to predict along-strike, updip, and downdip, or vertical changes in the dominance of depositional processes acting at any given location through geologic time. The key implications of these models are that given sets of known parameters, dominant and subordinate depositional

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processes or ranges of potential dominant and subordinate depositional processes acting at a coastline can be predicted. This provides an auditable methodology for determining reservoir modeling scenarios and reducing and managing the uncertainties in predictions of changes in clastic coastal depositional processes through time and space.

INTRODUCTION

The ability to predict the products of depositional processes acting along clastic coastlines (Figure 1) is an important aspect in the exploration for and the exploitation of hydrocarbons. Consequently, these depositional environments have been the subject of study, classification, and model building for 100 yr or more (Barrell, 1912; Bruun, 1962; Swift, 1968; Reading, 1978; Walker, 1979; Clifton, 2006). It is well understood that depositional processes acting at a shoreline determine the geometry and architecture of both sand bodies and potential heterogeneities (Flint and Bryant, 1993; Reynolds, 1999; Howell et al., 2008). Consequently, depositional processes are first-order controls on factors such as sand-body connectivity and stratigraphic compartmentalization in the subsurface (Ainsworth, 2005; 2010).

Current depositional models for clastic coastal systems are relatively simple and commonly focus on the dominant process that acted at the time of deposition. The most widely used models for marginal marine systems were developed in the 1970s by Coleman and Wright (1975) and Galloway (1975), who focused on and classified modern deltas using a ternary process-based approach anchored to the relative influences of waves, tides, and fluvial input. Boyd et al. (1992; 2006) expanded this ternary approach to include other types of non-fluvial coastal depositional systems. They linked the evolution of coastal systems to progradational and retrogradational shoreline episodes and, hence, introduced an element of prediction into their models. Such models, however, deal poorly with dynamically changing shallow-marine systems through time and space and tend to be descriptive instead of predictive. That is, they describe the products of processes acting at shorelines instead of predicting the type of coastal depositional processes that will be encountered under changing external forcing conditions either in space (along depositional strike or updip and downdip) or through time. In addition, these models do not take account of the potential amplifying or moderating effects of coastal morphology on the depositional processes or effects linked to fluctuations in rates of accommodation development

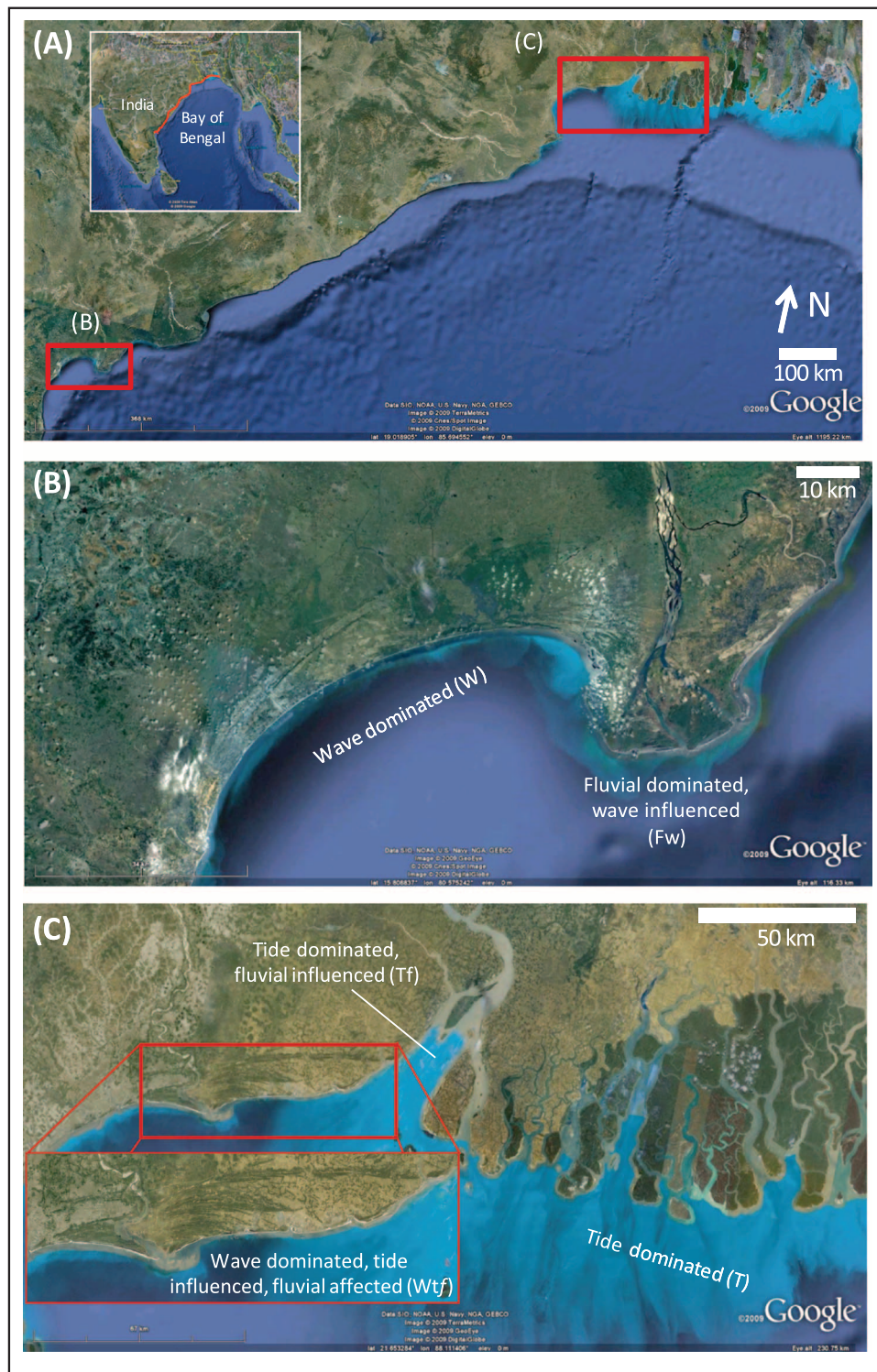


Figure 1. Impact of changing depositional processes on sand-body geometries. Examples from the Bay of Bengal (A) that illustrate how changing depositional process dominance results in different sand-body and coastal geometries (B and C) and also how dominant processes can change laterally (along depositional strike) over relatively short distances. Images courtesy of Google Earth.

and sediment supply, shoreline trajectory, or shelf width. Ainsworth (2003) and Ainsworth et al. (2008) built on the previous process-based models and integrated accommodation, sediment supply, and coastal morphology into a predictive matrix for

coastal depositional processes acting at a shoreline. However, they did not fully consider the impact of shelf width or overall basin shape on primary, secondary, and tertiary depositional processes acting at the shoreline.

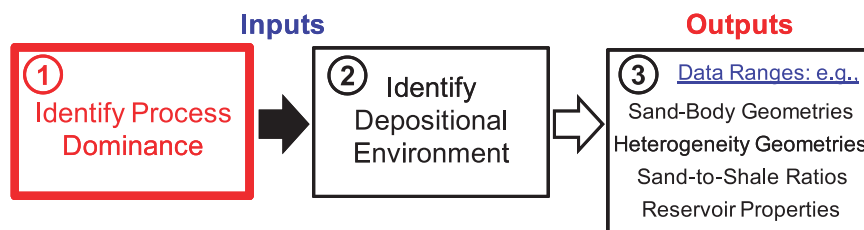


Figure 2. Process to product work flow for the collation and derivation of data for coastal depositional environments. The initial step of identifying process dominance is the subject of this article (1). A secondary step is required to identify depositional environments (2). Once process dominance and depositional environment are determined, data ranges for parameters such as sand-body and heterogeneity geometries, sand-to-shale ratios, and reservoir properties can be collated and stored in databases for subsequent usage (3). For example, the dominant depositional processes observed in core may be interpreted as fluvial dominated, tide influenced (1). These rocks may then be interpreted as being part of a mouth-bar depositional element (2). Data on geometries and properties can now be measured and stored in a database (3). Alternatively, if data output is required, the database can be queried to extract geometry and property data on fluvial-dominated, tide-influenced mouth-bar elements (3).

Work on modern systems (Anthony and Orford, 2002; Bhattacharya and Giosan, 2003; Sidi et al., 2003; Anderson and Fillon, 2004; Yang et al., 2005; Dalrymple and Choi, 2007; Hampson et al., 2008b) suggests that many present-day shoreline systems are relatively complex and are composed of discrete arrangements of depositional elements (Figure 1). These individual elements can be related to the relative importance of waves, tides, and fluvial processes, which in turn can vary through time, on daily, seasonal, annual, or longer time scales. The altering balances of these three processes through time result in many systems being significantly influenced by one, two, or even three of the processes at the same time. These systems are known as mixed-process systems. These types of systems have been recognized in the ancient record (Ainsworth, 2003; Willis, 2005; Yoshida et al., 2007; Ainsworth et al., 2008; Charvin et al., 2009); however, the complexity described in both the modern and the ancient is poorly reflected in existing coastal depositional system models for the ancient. Furthermore, no available consistent classification scheme exists to define or name these mixed-process systems. Recent improvements in the understanding of the evolution of these types of systems (Yoshida et al., 2007; Hampson et al., 2008b) and the impact of coastal morphology or paleogeography on depositional style (Ainsworth, 2003; Ainsworth et al., 2008) suggest that existing classifications and models can be further developed and improved. Indeed, the ternary diagrams of Galloway (1975)

lend themselves to a semiquantitative approach to classification of these mixed-process systems. Until now, this semiquantitative approach has not been attempted.

Models for coastal depositional systems commonly incorporate the processes that operate along shorelines (wave, tide, and fluvial) and relate these processes to a resultant depositional environment and sedimentary architecture (Figure 2). This article will address the first part of this work flow, the classification and prediction of depositional processes acting along clastic shorelines. This first step provides a framework within which the second step of linking the impacts of the resulting combination of processes and physiographic settings on depositional environments, sand-body, and heterogeneity geometries can be applied (Figure 2). This linkage of process and product is the subject of ongoing research (Vakarelov et al., 2010).

The objectives of this article were threefold:

1. To provide a new semiquantitative process-based classification scheme for coastal depositional systems that can be used to classify both simple shorelines and complex mixed-process systems. The classification scheme is intended to provide a pragmatic framework for the further development of studies of clastic coastal depositional systems and to enable the direct comparison of both modern and ancient shoreline deposits by providing a common and consistent process-based approach.

2. To further develop predictive clastic shoreline process models by integrating the impacts of coastal morphology, accommodation, sediment supply, shelf width, and basin geometry with the aim of improving the dynamic predictive capabilities of these models through time and space. The application of these concepts is illustrated by worked examples.
3. To provide a methodology with a simple audit trail for the prediction of depositional process dominance along clastic shorelines. This will improve our ability to manage and reduce uncertainties in the prediction of dominant and subordinate processes along ancient clastic shorelines. It will also provide a framework that will enable reduction in uncertainties in the prediction of depositional environments, types, geometries, and properties of reservoir sand bodies and heterogeneities, and improve our ability to manage these uncertainties.

EXISTING DEPOSITIONAL MODELS FOR CLASTIC COASTAL SYSTEMS

Deltas or fluviatile coastlines have long been a focus of research because they host numerous major hydrocarbon accumulations around the globe (Ahlbrandt et al., 2005). Although the focus of hydrocarbon exploration and production has been on deltas, a large proportion of clastic coastlines around the world are nonfluviatile. The subsequent section briefly details the evolution of models of both fluviatile and nonfluviatile coastlines.

Fluviatile Coastlines: Deltas

Deltas were the first focus of recent depositional model building for clastic coastal systems. Coleman and Wright (1975) used a statistical approach to categorize the morphology of modern systems; Galloway (1975) developed a ternary diagram based on the relative influences of rivers, waves, and tides on deltaic systems, which can be classified as fluvial, wave, or tide dominated. This classification has stood the test of time and adequately describes deltas that encompass the end members of fluvial,

wave, or tide dominance. These authors also noted that many modern deltas show evidence of mixed processes where two or three of the processes exhibit a significant influence on the morphology of the delta. Since the mid-1990s, several detailed studies of these mixed-influence deltas have been published. Bhattacharya and Giosan (2003), for example, detailed a series of modern wave-dominated deltas that show significant fluvial influence in which wave-dominated updrift parts can differ significantly from fluvial-dominated downdrift parts. The mixed-influence motif of some deltas has also been recognized in ancient systems. Numerous authors have noted vertical, downdip and along-strike changes in depositional process dominance in ancient deposits (Mellere and Steel, 1995; Bhattacharya and Giosan 2003; Barton et al., 2004; Gani and Bhattacharya., 2005; Ainsworth et al., 2008; Plink-Bjorklund, 2008). Bhattacharya and Walker (1992), Giosan and Bhattacharya (2005), and Bhattacharya (2006) provide excellent summaries and examples of deltaic depositional models.

Nonfluviatile Coastlines

Recent models for nondeltaic coastlines were developed in the late 1980s by Boyd et al. (1992, 2006), who built on the Galloway deltaic ternary plots, and a wave and tide coastal classification by Hayes (1975, 1979). They introduced a predictive element by considering the likelihood of certain depositional environments to occur, depending on whether a system is transgressive or regressive. They suggested that during regressions, deltas, strand plains, cheniers, and tidal flats form preferentially, whereas during transgressions, estuaries and barrier-island systems are more likely to be generated.

Yoshida et al. (2007), building on work by Cram (1979) and Howarth (1982), suggested that predictable changes occur as depositional systems transit across shelves during transgression and regression. Wave processes are assumed to dominate at the shelf margin during lowstands in sea level as nothing exists to impede waves striking the shoreline and nothing exists to amplify the tidal wave as it approaches the shoreline. During transgressions and highstands, however, a wider shelf

increases the importance of tidal processes because of increased tidal resonance; this is commonly coupled with a corresponding dampening of wave power caused by frictional energy losses to the sea floor. The opposite is suggested to be true during regression with changes from tide domination at highstand back to wave domination at lowstand shelf margins.

Ainsworth (2003) and Ainsworth et al. (2008) coupled the impacts of coastal morphology with the concepts of accommodation-to-sediment supply (A/S) ratios to predict the occurrence of primary and secondary processes in mixed-process systems along both fluvial and nonfluvial shorelines in settings where tidal influence was predominantly driven by shoreline morphology. This article further develops these concepts to incorporate the potential impact of shelf and basin morphology on tidal influence at the shoreline. It also expands the predictive matrices of Ainsworth (2003) and Ainsworth et al. (2008) to include the prediction of primary, secondary, and also tertiary process dominance.

COASTAL PROCESS CLASSIFICATION

Process classification is an important first step to develop a consistent conceptual framework for subsequently predicting processes and then analyzing depositional environments and depositional elements (Figure 2). Dominant depositional processes are readily identifiable in modern and ancient systems, and coastlines are routinely classified as wave, tide, or fluvial dominated. Mixed-process fluvial and nonfluvial systems are also being increasingly recognized in both modern and ancient systems (Hampson et al., 2008a). However, no available consistent classification scheme exists to define or name these mixed-process systems in a rigorous auditable manner. Neither is a mechanism available for the prediction of changes in the dominance of processes along these coastlines through time and space. This section defines a new semi-quantitative classification scheme for process dominance along clastic depositional coastlines. The

following sections of the article define a simple auditable methodology to predict changes in process dominance through time and space.

Full Classification Concept

Panels A and B in Figure 3 illustrate the proposed full classification system for clastic coastal systems. It is an expansion of the ternary plot first used by Galloway (1975) for fluvial coastlines (deltas) and has been expanded to include nonfluvial coastlines that occur on the bottom side of the triangle that shows only wave and tidal influence (Figure 3A). The classification uses discrete categories that are based on a dominant process, as well as a secondary and tertiary process clarifiers. For example, on the bottom side of the triangle in Figure 3A, a nonfluvial coastline can be wave dominated (W); wave dominated tide influenced (Wt); wave and tide influenced (when wave and tidal processes are equal [wt]); tide dominated, wave influenced (Tw); or tide dominated (T). When this approach is expanded to include fluvial influence, an additional 17 different combinations of fluvial, tidal, and wave processes are introduced (Figure 3A). Where three processes are involved, the most important process is said to “dominate,” the second most important process is said to “influence,” and the third most important process is said to “affect” deposition. For example, a system may be classified as fluvial dominated, tide influenced, wave affected (Ftw). Such an approach allows a robust categorization of complex natural systems, whereby subtle differences in mode of deposition can be easily communicated.

Simplified Classification Concept for Prediction of Coastal Processes

Panels A and B in Figure 3 show the full spectrum of possible combinations of wave, fluvial, and tidal processes and detail 22 potential classification categories. In a practical sense, this is a large number of categories to handle. To simplify the classification, some categories can be removed from the full scheme

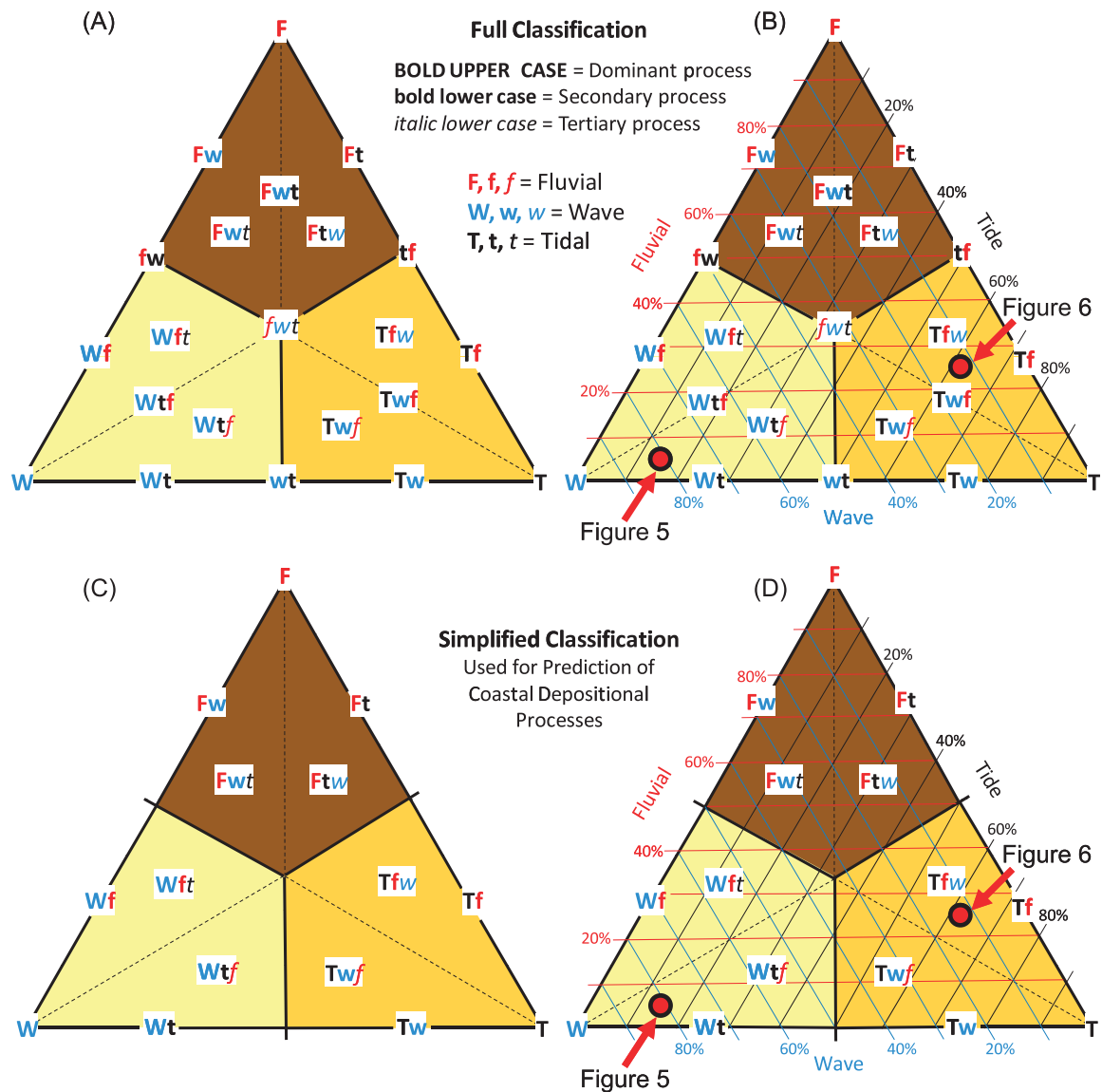


Figure 3. Coastal process classification ternary plots modified from Galloway (1975). The bottom side of the triangles represent classification of nonfluvial shorelines. All other portions of the plots represent some degree of fluvial influence. Panels A and B illustrate the full classification of dominant, secondary, and tertiary processes. Panels C and D represent a simplified classification. See Figure 4 for representative coastal geometries associated with these fifteen simplified classification categories. This simplified classification is used for the predictive matrices (Figure 9) and decision trees (Figure 10) used in this article. Panels B and D illustrate how percentages of sedimentary structures from cores (Figure 5) or areal coverage from modern systems (Figure 6) can be plotted on the diagrams. F = Fluvial dominated; W = Wave dominated; T = Tide dominated; Fw = Fluvial dominated, wave influenced; Ft = Fluvial dominated tide influenced; Tf = Tide dominated fluvial influenced; Tw = Tide dominated wave influenced; Wt = Wave dominated tide influenced; Wf = Wave dominated fluvial influenced; Fwt = Fluvial dominated, wave influenced, tide affected; Ftw = Fluvial dominated, tide influenced, wave affected; Tfw = Tide dominated, fluvial influenced, wave affected; Twf = Tide dominated, wave influenced, fluvial affected; Wtf = Wave dominated, tide influenced, fluvial affected; Wft = Wave dominated, fluvial influenced, tide affected; fw = fluvial and wave influenced; tf = tide and fluvial influenced; wt = wave and tide influenced; fwt = fluvial, wave and tide influenced; Fwt = Fluvial dominated, wave and tide influenced; Twf = Tide dominated, wave and fluvial influenced; Wtf = Wave dominated, tide and fluvial influenced.

without significantly hindering the effectiveness of the classification. Categories where the influence of each process is identical (wt, tf, fw, fwt; Figure 3A)

have a very low probability of occurrence and can therefore be readily removed (Figure 3C). Categories where the secondary and tertiary processes

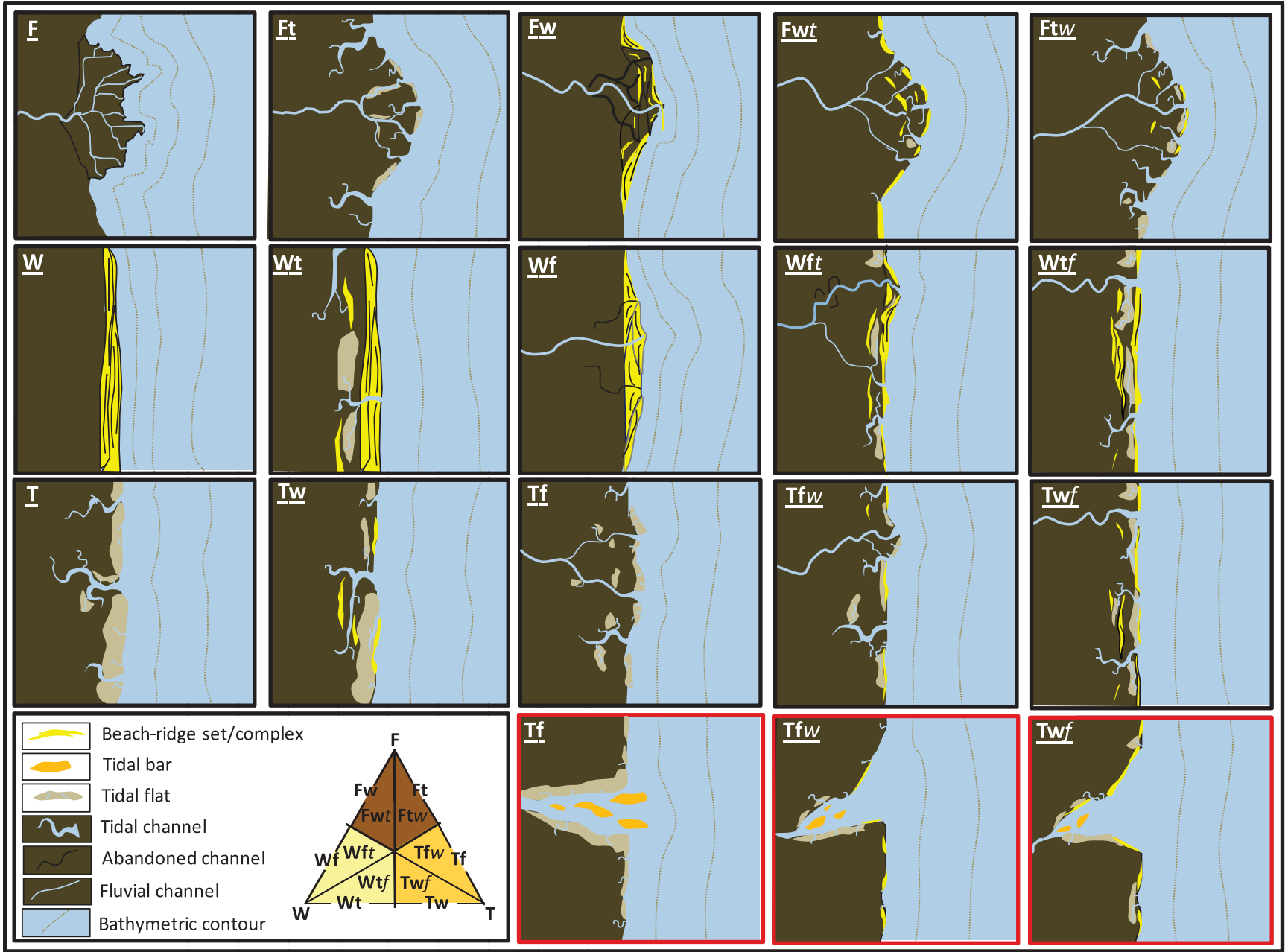


Figure 4. Representative schematic plan view models of the 15 classification categories in the simplified coastal process classification presented in Figure 3C and D (see text and Figure 3 caption for explanation of classification category labels). The figure was developed from analysis of architectures of modern systems. Note differences in geometries and depositional elements in the different categories. Red boxes around the Tf, Tfw, and Twf (bottom right) represent differences in architecture of these categories when part of an embayed shoreline. Categories T and Tw can occur along stretches of shoreline within the embayed Tf, Tfw, and Twf categories. Map scales and labels of bathymetric contours have purposefully been omitted to keep the models generic.

are equal (Fwt, Twf, Wtf; Figure 3C) can also be reasonably removed as they will also have a very low probability of occurrence.

The result of the above simplifications is the “simplified classification” shown in Figure 3C. In this figure, 15 possible combinations or categories of shoreline depositional process combinations exist. Figure 4 is derived from observations of modern systems. It illustrates schematic but representative plan-view geometries of systems in each of the 15 categories. Note the distinct differences in coastal geometries that result from the different combinations of processes.

Application of Classification to Ancient Systems

In this classification scheme, depositional systems are initially defined based on the relative influence of tidal, wave, and fluvial processes that acted on the sediments as they were deposited. In the ancient, the assumption is that the dominant depositional processes acting at the time of deposition would have produced the largest proportion of sedimentary structures that will hence have had a greater chance of preservation in the rock record. Therefore, a preserved succession that has predominantly wave-generated sedimentary structures, with a subordinate percentage of tide-generated structures and even fewer fluvial-generated structures (Figure 5) would be classified in Figure 3 as a wave-dominated, tide-influenced, fluvial-affected system (Wtf). If the exact percentage of sedimentary structures generated by each process can be calculated from core or from outcrop (Figure 5), the system can be plotted on the ternary plots (Figure 3B, D) and the classification can be determined. Note that a degree of interpretation is sometimes required to attribute sedimentary structures to their formative process. Some sedimentary structures are diagnostic of process; for example, hummocky and swaley cross-stratification are wave-generated structures. However, other sedimentary structures can be formed by two or more processes; for example, trough cross-stratification could be formed by waves in the upper shoreface or by fluvial-generated currents in a distributary

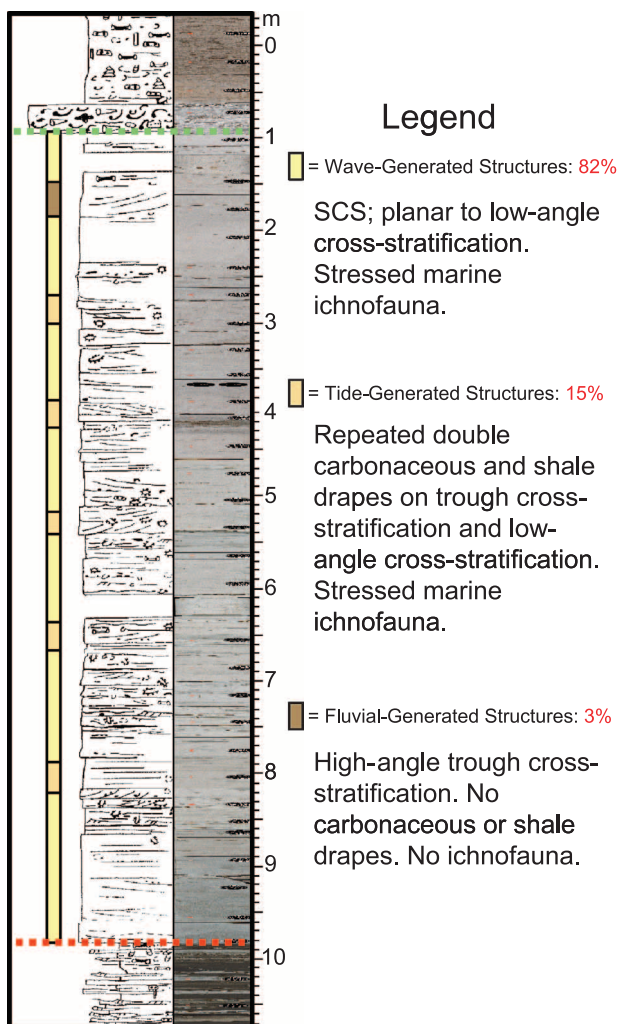


Figure 5. Ancient process classification example. Section of core from well Sunrise-2, Timor Sea, Australia (Ainsworth, 2003), in which the sedimentary structures have been attributed to being generated by wave processes (82%), tidal processes (15%), and fluvial processes (3%). The interval between the red and green hatched lines is therefore classified as wave dominated, tide influenced, fluvial affected (*Wtf*; see Figure 3). SCS = Swaley cross-stratification.

channel. In these cases, other lines of evidence such as associated and adjacent sedimentary structures, paleogeographic setting, and ichnological data (cf. MacEachern and Bann, 2008) must be used to interpret the process that generated the structure. In some cases, the interpretation will be equivocal, and this will represent the inherent uncertainty present in the data set. Steel et al. (2008) provide a guide to the recognition of fluvial, tidal, and wave processes in rocks deposited in marginal marine settings.

Application of Classification to Modern Systems

When examining modern systems, similar concepts can be applied. If depositional elements can be identified as being deposited by a dominant depositional process, then the area covered by each element can be calculated and the corresponding relative impact of each depositional process can also be determined. Figure 6A shows an application of this approach to the Mitchell River Delta in the Gulf of Carpentaria, Australia. The depositional elements in this system have been mapped and dominant processes have been assigned to each element (Massey, 2009). Tide-dominated elements cover 62% of the area, fluvial-dominated elements cover 25% of the area, and wave-dominated elements cover 13% of the area. On the classification schemes in Figure 3B and D, this system would plot as a tide-dominated, fluvial-influenced, wave-affected system (*Tfw*).

Validity of Concepts

As with any classification scheme, a basic assumption for applying these percentage calculation concepts are that the areas, or vertical sections studied, are representative of the system being classified. This is a particular issue when dealing with sub-surface or outcrop data where only restricted windows to the world are available. The black dots numbered 1 to 3 on Figure 6A illustrate this point. A well at 1 would suggest wave domination of the system. A well at 2 would suggest fluvial domination of the system. A well at 3 would suggest tidal domination of the system. While these interpretations will be correct for the local areas being studied, they will be incorrect for the system as a whole. Therefore relatively small amounts of data will lead to a greater uncertainty in the interpretations of those data. The greater number of cored wells in an area will therefore lead to a greater confidence in the interpretation of the process dominance of the whole depositional system being correct.

By the same degree, a limited outcrop area such as the area illustrated by the red box at location 4

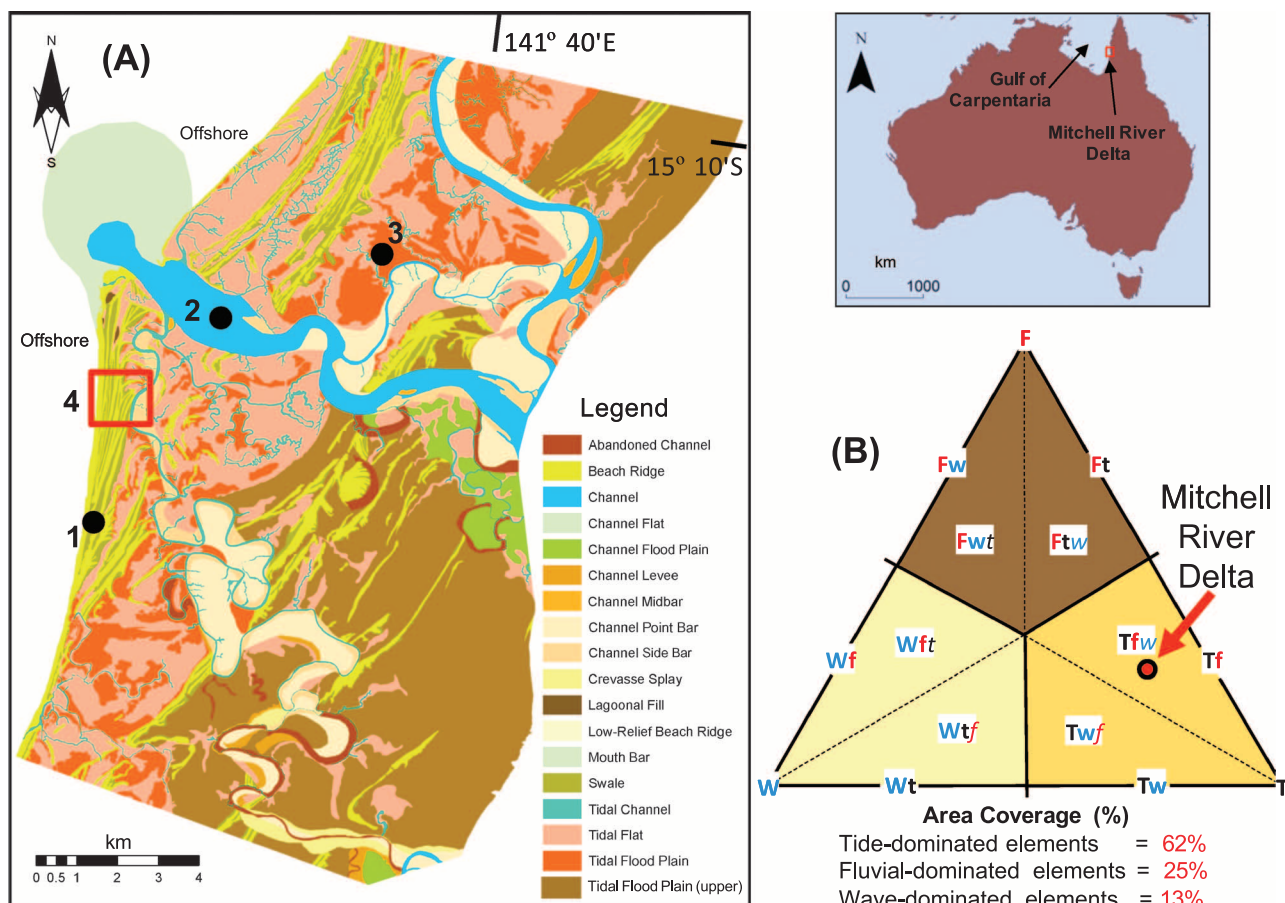


Figure 6. Modern process classification example. Mitchell River Delta, Gulf of Carpentaria, Australia (location map inset). (A) Depositional environments have been classified from remote sensing and fieldwork and attributed to dominant depositional processes (Massey, 2009). The percentage of the area of the whole system covered by elements generated by tidal (62%) fluvial (25%) or wave processes (13%) was calculated and plotted on the simplified ternary plot (B). It is therefore classified as a tide-dominated, fluvial-influenced, wave-affected system (Tfw; see Figure 3). Numbers 1 to 3 on panel A represent the location of potential wells that would give different process dominance results to the whole system if the well results were analyzed individually. (1) would be wave dominated, (2) fluvial dominated, and (3) tide dominated. Number 4 on panel A represents a potential outcrop area, which would again give a different process domination result from the whole system analysis; see text for discussion. Panel A is modified from Massey (2009).

on Figure 6A would probably lead to an interpretation of a wave-dominated tide-influenced system (Wt). Although this is correct for the area studied, it is not the correct interpretation for the system as a whole. Outcrop areas therefore suffer the same potential classification issue if the objective is to classify the whole depositional system.

These principles of percentage of rock generated by different processes can be applied to rock units on any scale. For example, these calculations can be made on a parasequence, systems tract, sequence, or sequence set scale (Ainsworth et al., 2008). By the same degree, if modern systems were

analyzed, parts of coastlines, depositional systems, or even continental margins could be classified using these percentage concepts.

An argument that is commonly leveled against analyzing modern coastal systems (Figure 6) is that the preservation potential of modern systems is low. If a modern coastal system is deltaic, a possibility exists that during long periods of progradation, fluvial systems may rework previously deposited wave, tidal, or fluvial deposits in the coastal plain area. However, as can be seen from the mapping in Figure 6A, the elements described for this delta have already been preserved. A rapid

rise in relative sea level with negligible transgressive erosion could lock in the present-day sedimentary architecture for preservation in the rock record. The study of modern examples and their preserved architectures is thus very relevant as potential analogs applicable to the ancient and to the subsurface.

COASTAL DEPOSITIONAL PROCESS PREDICTION WITH LIMITED DATA

Using the simplified classification scheme previously developed (Figure 3C, D), this section builds on and further develops the predictive models for coastal depositional process dominance developed by Ainsworth (2003) and Ainsworth et al. (2008). The text below therefore summarizes the concepts and applications detailed by these publications, and the reader is referred to these publications for more detailed explanations of the concepts. Application of these concepts in the modern is relatively simple because most of the required parameters can be directly observed. However, in the ancient, the definition of these parameters is more problematic. The subsequent text, related to application of the concepts, therefore focuses on how they can be applied to the ancient.

Four key parameters are convolved to predict primary, secondary, and tertiary depositional processes acting at the shoreline; the likelihood of wave versus fluvial effectiveness; the influence of coastal morphology; the influence of shelf width or basin geometry on tidal resonance potential; and the influence of A/S ratio.

Key simplifying assumptions with these models are that climate and paleolatitude remain constant during the period being predicted. Changes in the caliber of sediment input during the period predicted are also not considered by the models. Arguably, the actual physical process domination at the shoreline can be said to be independent of the caliber of sediment supplied to the coast. However, the facies and architecture of sediments deposited during periods of varying sediment caliber supply will potentially be very different (cf. Orton and Reading, 1993; Yoshida et al., 2007). The key

impacts of climate change during a depositional period may be changes in sediment caliber (see previous discussion) and also differences in absolute volumes of coarse sediment delivered to the coastline. The volume of coarse sediment will be reflected in the accommodation to the sediment supply ratio parameter, which is subsequently considered in the section titled Influence of Accommodation and Sediment Supply.

Wave Versus Fluvial Effectiveness

Concept

This parameter is simply related to the balance of wave and fluvial energy in a system. If a system is affected by a large river that has the ability to transport sediment to the coastline at a rate greater than any waves present can redistribute that sediment, then the area is said to have high fluvial effectiveness and low wave effectiveness. Conversely, if the waves are able to redistribute the sediment supplied by the river more quickly than it is transported to the coastline, then the system can be said to have high wave effectiveness and low fluvial effectiveness. If no rivers are present, then a system will a priori have relatively high wave effectiveness.

Application

The determination of high versus low wave and fluvial effectiveness in the ancient relies on knowledge of the paleogeography of an area. If an area of coastline is open to a large body of water or an ocean, a high probability exists that it will have a high wave effectiveness. However, if a body of water is sheltered from oceanic waves by a headland, an island, or by rugose coastal morphology, then it may be considered to have a probability of high fluvial effectiveness and low wave effectiveness. If evidence for ancient rivers can be found in the rock record, then the area can be said to potentially have relatively high fluvial effectiveness. However, fluvial process dominance in these circumstances is not always the case as wave processes can still redistribute sediment supplied by the rivers, and hence waves can become the dominant depositional process (e.g., Wf, Wft; and Wtf; Figures 3, 4). Alternatively, if paleogeographic

information is available as to the size of the fluvial drainage areas, predictions of wave versus fluvial effectiveness can also be made on this basis. Relatively large drainage basins would have a higher probability of producing high fluvial effectiveness at the coastline.

Influence of Coastal Morphology

Concept

Coastal morphology can have very important effects on the processes acting at a shoreline (Ainsworth et al., 2008). With regard to tides, highly embayed or funnel-shaped coastal morphologies have the potential to amplify the tidal wave as it moves into an area, thereby increasing the potential for increased tidally generated current velocities and the transport of sediment because of tides (Wells et al., 2007). The morphology of the coastline is hence used as a proxy for tidal influence, with increasing coastal rugosity considered to increase the probability of tidal influence at the shoreline.

A secondary impact of coastal morphology relates to waves and rivers. With regard to waves, straight to lobate shorelines provide no protection from oceanic waves and are hence likely to be open to the full force of any wave processes. More complex moderately to highly embayed shorelines have the potential to shelter the coastline from direct wave approach and may, depending on the wave approach direction relative to the orientation of any headlands, result in more protected coastal settings and relatively lower wave energy being experienced at the shoreline. In these settings, protected from direct wave approach, any fluvial systems discharging into the ocean are also likely to have a larger degree of fluvial effectiveness relative to the wave energy.

Application

Definition of coastal morphology in the ancient relies on knowledge of the paleogeography of an area. If an area of the shoreline is represented by moderately to highly embayed morphologies, then a high probability exists that the tidal wave will be amplified by these morphologies and that tidal

processes will be significant in the area. Similarly, the rugosity of a coastline may also serve to protect the shoreline from oceanic waves and reduce the wave effectiveness with respect to fluvial effectiveness. If, however, an area of coastline is straight to lobate and open to a large body of water or an ocean, a high probability exists that it will not be affected by increased tidal activity, unless it is also associated with a wide shelf (see following section).

Influence of Shelf Width or Basin Geometry on Tidal Resonance Potential

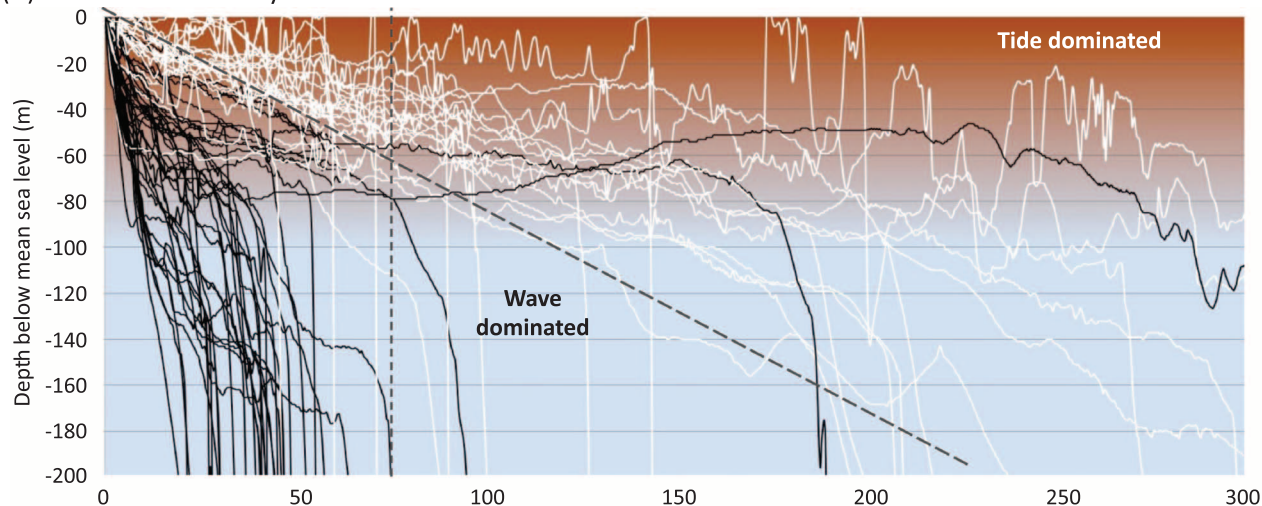
Concept

This is a new component added to the previous work of Ainsworth (2003) and Ainsworth et al., (2008). The impact of shelf width and basin geometry on the tidal resonance potential of a basin was not fully considered by these authors. Hence, these earlier models are only applicable to coastlines along basins that have a low tidal resonance potential generated by the basin morphology. These earlier models relied on coastal morphology to amplify the tidal wave as it moved into a coastal area (see 2). However, some areas already have a relatively high potential for high tidal ranges and effective tidal prisms simply because of the amplifying effect of the shelf width or the basin morphology on the tidal wave (Cram, 1979; Howarth, 1982; Pugh, 1987).

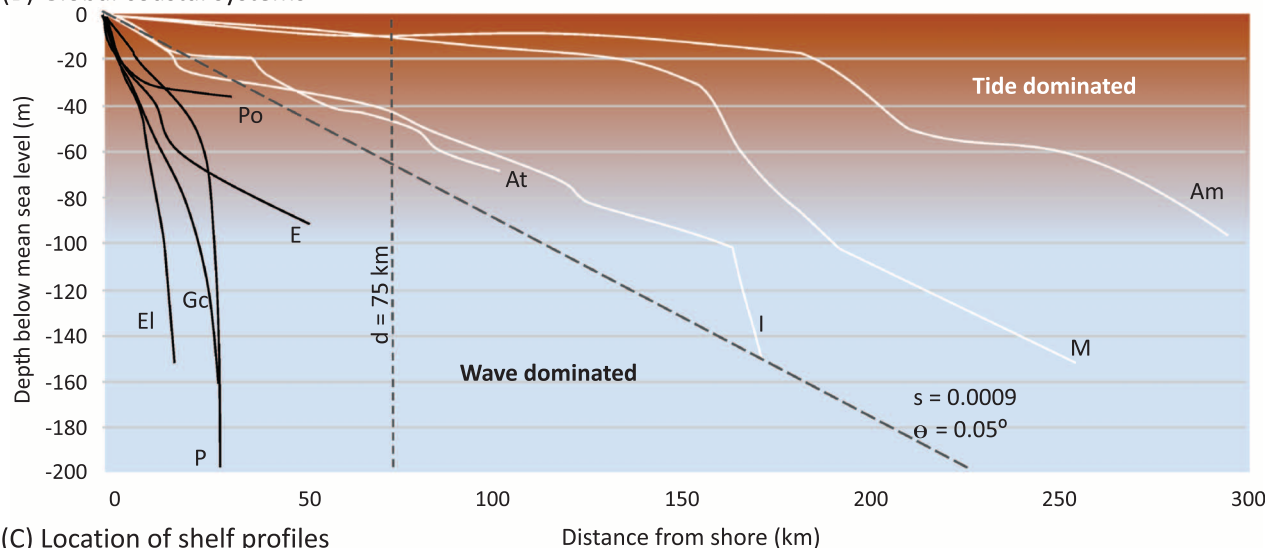
Application

Increasing shelf width is a well-known factor for the potential amplification of the tidal wave by resonance effects (Howarth, 1982). However, to apply this parameter to the ancient, an empirical cutoff shelf width value is required to differentiate between high and low tidal resonance potential shelves. Figure 7A illustrates a series of shelf profiles taken from around the coastline of Australia. The white lines represent shelves whose corresponding shoreline is classified as tide dominated (Heap et al., 2004). The black lines represent shelves whose corresponding shoreline is classified as wave dominated (Heap et al., 2004). A noticeable change is seen in process dominance at an approximate

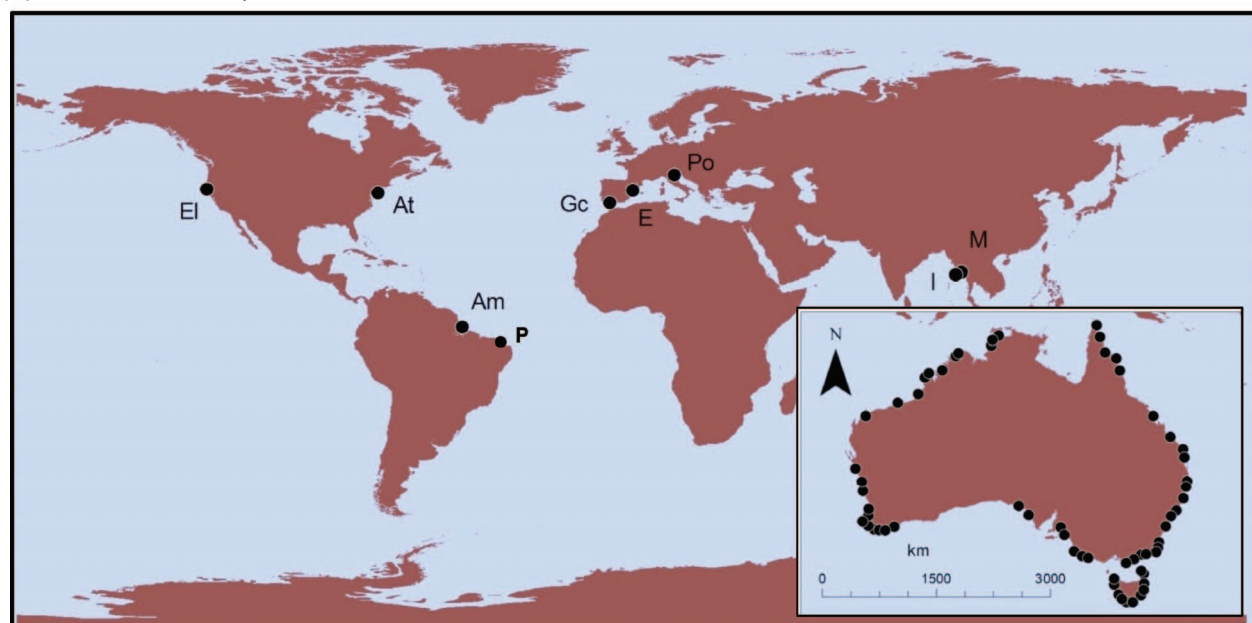
(A) Australian coastal systems



(B) Global coastal systems



(C) Location of shelf profiles



shelf width of 75 km (47 mi) from wave dominated (shelf width <75 km [<47 mi]) to tide dominated (shelf width >75 km [>47 mi]). To test this shelf width value against other coastlines around the globe, nine other geographically diverse locations were selected and plotted on the same scale (Figure 7B). These examples also suggest that a 75-km (47-mi) shelf width is significant. For the predictive models, we therefore suggest that a shelf width of less than 75 km (<47 mi) is likely to have a low potential for tidal resonance and hence generating tidal currents at the shoreline that are related to amplification of the tidal wave by the shelf width itself. By the same degree, we also suggest that shelf widths more than 75 km (>47 mi) have a relatively high potential for tidal resonance, and hence generating tidal currents at the shoreline that are related to amplification of the tidal wave by the shelf width itself.

Prediction of the impact of basin morphology on tidal ranges and tidal prisms at the shoreline is a complex issue (Mitchell et al., 2010). However, general acceptance exists that basins that have restricted access to the open oceans do have a relatively low potential for generating tidal currents by amplification of the tidal wave because of resonance effects (Ericksen and Slingerland, 1990; Wells et al., 2005a, b, 2007; Mitchell et al., 2010). In these types of settings and on shelves less than 75 km (<47 mi) wide, tidal influence at the shoreline depends most strongly upon the morphology of the coastline itself and the ability of that morphology to amplify the tidal wave (see 2).

On the contrary, certain restrictive basinal geometries exist such as narrow straits that may focus, funnel, and amplify tidal energy (Leckie and Rumpel, 2003; Mitchell et al., 2010). Such geometries may commonly be found in tectonically extensional settings. The main driver for tidal am-

plification in these settings is caused by the basin geometry, and for the purposes of this model, they can be considered to effectively have a high tidal resonance potential.

Influence of Accommodation and Sediment Supply

Concept

The A/S ratio principles have been well documented by many authors (Cross 1988; Bhattacharya and Walker, 1992; Boyd et al., 1992; Cross et al., 1993; Schlager 1993; Muto and Steel, 2002; Barton et al., 2004; Gardner et al., 2004). The A/S ratios represent the rate of change of accommodation versus the rate of change of sediment supply. As a basic concept, the ratio of A/S does not directly influence the depositional processes acting at a shoreline (Ainsworth et al., 2008). However, two key circumstances are present, where A/S can be very important.

1. When A/S is combined with rugose coastal morphology, systems can switch from being wave dominated to fluvial dominated. In highly to moderately embayed settings under high A/S conditions, the rate of generation of accommodation space may be too great for the rate of sediment supply to infill. The embayed nature of the shoreline will therefore remain and potentially protect the area from direct wave approach (Ainsworth, 2003; Ainsworth et al., 2008). This may result in a predominance of high A/S systems in embayed settings having a relatively low wave effectiveness.
2. A second example of where A/S may be important in determining process dominance at a shoreline may be in areas that have a high tidal

Figure 7. Shelf width versus coastal process dominance. Shelf profiles orientated perpendicular to (A) the Australian coastline and (B) global coastline examples. Coastal systems interpreted as tide dominated are represented by white lines and wave-dominated systems by black lines. Wave-dominated systems tend to be located where shelves are less than 75 km (<47 mi) in width and where shelf slope gradients are greater than 0.0009. Where shelf widths exceed 75 km (47 mi) in width and slope gradients are less than 0.0009 tidal processes dominate at the coastline. Coastal dominance interpretation data in panel A is from Heap et al. (2004). Panel C illustrates point locations for each of the shelf profiles. El = Eel River; At = Atlantic City; Am = Amazon River; P = Pacatuba; Gc = Gulf of Cadiz; E = Ebro; Po = Po Delta; I = Irrawaddi; M = Martaban.

resonance potential because of a relatively wide shelf (see 3). Tide domination is the norm for low A/S regimes on wide shelves (Hubbard et al., 1979). However, on straight to lobate shorelines in these settings when high A/S conditions are encountered, the high rate of accommodation development may allow other processes to dominate. In high wave-effectiveness regimes, a high rate of accommodation development would result in reduced frictional dampening of waves across the shelf as well as a potential for reduced tidal wave amplification because of increasing water depths. This would hence result in a higher potential for wave dominance at the shoreline. In low wave-effectiveness regimes, if rivers are present during high A/S episodes, the potentially reduced tidal wave amplification caused by increasing water depths may allow fluvial processes to become dominant over tidal processes.

In addition, the geometries of sand bodies and heterogeneities influenced by the same depositional processes but deposited under high and low A/S conditions can potentially be very different (Enge and Howell, 2010; Ainsworth, 2010).

Application

The A/S ratio is a very useful theoretical but rather nebulous practical concept. Ratios less than 1 should represent progradation and ratios greater than 1 should represent transgression. Unfortunately, directly measuring either rates of change of accommodation or rates of change of sediment supply from an ancient succession is not possible.

As a practical differentiation, A/S ratios in this article are considered to be either low or high. Ratios of A/S can be related to parts of systems tracts (Ainsworth et al., 2008). Relatively low A/S ratios are typical of late highstand falling-stage and early lowstand systems tracts. Relatively high ratios of A/S are typical of late lowstand, transgressive, and early highstand systems tracts (Figure 8).

Shoreline trajectory concepts (Helland-Hansen and Martinsen, 1996; Helland-Hansen and Hampson, 2009) can also be used to determine whether a sys-

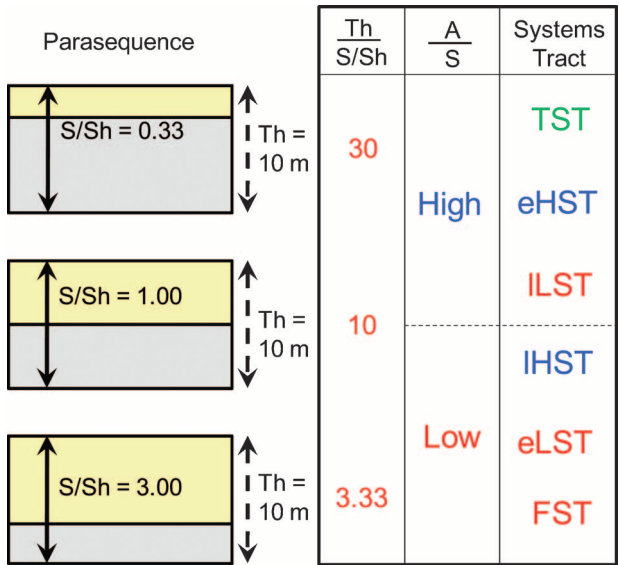
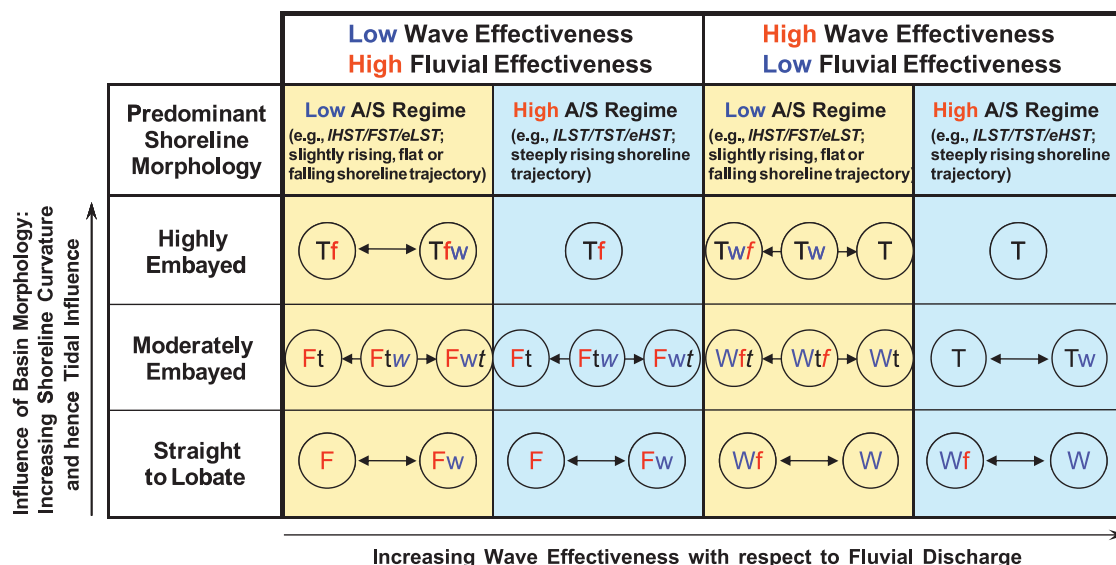


Figure 8. Accommodation/sediment supply (A/S) ratio, thickness/sand-to-shale ratio, and sequence stratigraphy. The A/S ratio can potentially be partially quantified by measurements of unit thickness (Th), which equates to accommodation, and sand-to-shale ratio (S/Sh), which equates to coarse sediment supply. These absolute values shown of Th divided by S/Sh ratio are indicative only. Further work is required to generate ranges of values that can be equated to relatively high and relatively low A/S regimes and perhaps to individual portions of systems tracts. TST = transgressive systems tract; eHST = early highstand systems tract; ILST = late highstand systems tract; IHST = late highstand systems tract; FST = falling-stage systems tract; eLST = early lowstand systems tract; ILST = late lowstand systems tract.

tem has a relatively high or low A/S ratio. Slightly rising flat or falling shoreline trajectories are related to low A/S settings. Steeply rising shoreline trajectories can be related to high A/S settings (Ainsworth et al., 2008).

Work by Ainsworth (2003, 2005) and Ainsworth et al. (2008) has also suggested that a proxy for A/S can be determined by measuring thickness of a sedimentary unit and dividing it by the sand-to-shale ratio of that unit. Accommodation is therefore approximated by thickness of a rock unit, and the coarse sediment supply is approximated by the sand-to-shale ratio of that unit. This assumes that relatively little of the sediment has been subsequently eroded from the unit. Figure 8 shows how this can be potentially applied quantitatively. Thickness divided by sand-to-shale ratios for units can be readily calculated from well or outcrop data and numbers documented. Figure 8 schematically

(A) **Matrix 1: Low Tidal Resonance Potential (e.g., Relatively Narrow Shelf <75 km)**
Tidal wave amplification by shelf geometry is negligible



(B) **Matrix 2: High Tidal Resonance Potential (e.g., Relatively Wide Shelf >75 km)**
Tidal wave amplification by shelf geometry is significant

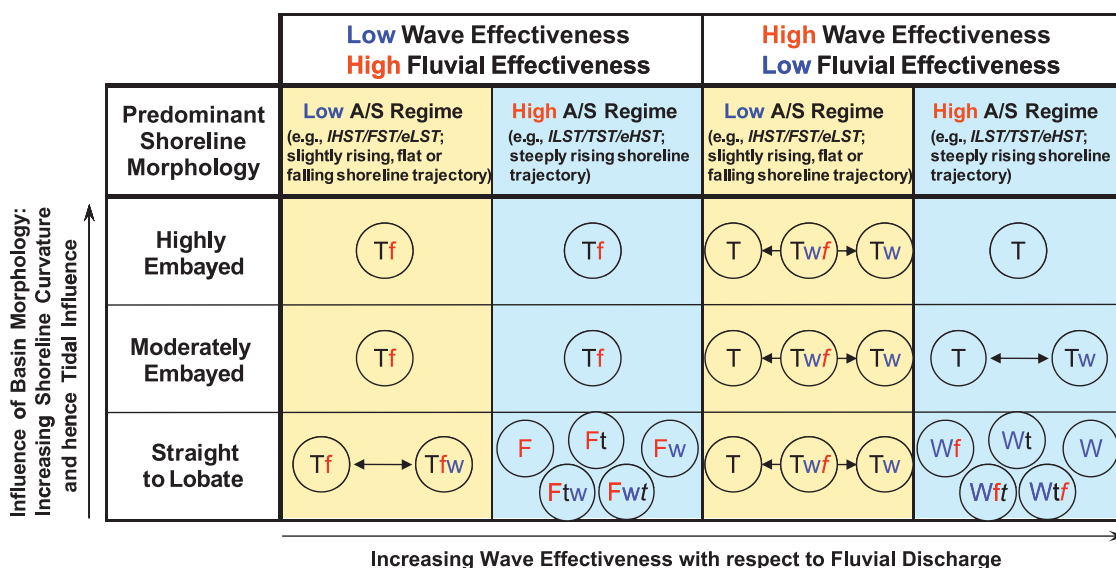


Figure 9. Matrices for prediction of depositional process dominance for clastic coastal systems. See Figure 3C and D for the simplified classification that these matrices are based on. (A) Matrix 1 is for settings with low tidal resonance potential (e.g., narrow, <75 km [<47 mi] shelves). Modified from Ainsworth et al. (2008). (B) Matrix 2 is for settings with high tidal resonance potential (e.g., wide, >75 km [>47 mi] shelves). In both panels A and B, on the high wave-effectiveness side of the diagrams, in high A/S regimes, and in highly and moderately embayed settings, waves are attenuated by shoreline morphology, thus allowing tidal processes to dominate. The matrices can also be represented as a decision tree (Figure 10). See Figure 3 for a definition of coastal process dominance acronyms. See Figures 11–13 for examples of how to use the matrices. See Figure 8 for sequence-stratigraphic abbreviations.

illustrates how these numbers may relate to high and low A/S regimes and their related systems tracts, but the numbers in this figure are for illus-

tration only. Further work is required to determine which exact ratios relate to relatively high and relatively low A/S regimes.

The Role of Sequence Stratigraphy

While sequence stratigraphy is an excellent method for subdividing stratigraphy into manageable and meaningful units (van Wagoner et al., 1988, 1990; Posamentier and Allen, 1999), it is unable to provide a guide as to the dominant or subordinate depositional processes acting at a shoreline (Ainsworth et al., 2008). This is mainly caused by the fact that processes acting at the shoreline are the product of more than just the accommodation and sediment supply parameters that are the basis of the sequence-stratigraphic approach. As previously described, processes acting at the shoreline are also products of parameters such as shelf width, basin morphology, and shoreline morphology. The sequence-stratigraphic approach does not take account of these parameters and is therefore unable to be used in a predictive process sense. However, certain systems tracts can be related to high and low A/S regimes (Ainsworth et al., 2008), hence, the systems tracts can be used as guides to the A/S setting. Sequence-stratigraphic interpretations can hence be incorporated into the predictive models (Figures 8, 9).

Predictive Matrices and Decision Trees

The four parameters previously detailed can be convolved and displayed as two matrices that can be used to predict primary, secondary, and tertiary processes acting along clastic coastlines (Figure 9).

Matrix 1 (Figure 9A) relates to basins with a low tidal resonance potential, that is, basins with relatively narrow shelves (<75 km [<47 mi] wide) or restricted access to open oceans. This model is expanded from the one proposed by Ainsworth et al. (2008), which only showed primary and secondary processes. In each box, ranges of potential process dominance are shown. For example, on a straight to lobate shoreline in a high wave effectiveness low A/S setting, process dominance is predicted to range from wave dominated (W) to wave dominated, fluvial influenced (Wf). With increasing shoreline embayment and approximately the same A/S, wave effectiveness is reduced (mov-

ing vertically on Figure 9A), and process dominance changes as tidal processes can now come into play, given the increase in tidal amplification by coastal morphology. Hence, processes can range from wave dominated, tide influenced (Wt) to wave dominated, tide influenced, and fluvial affected (Wtf), to wave dominated, fluvial influenced, and tide affected (Wft). Moving farther up the matrix to a highly embayed setting, tidal processes can now dominate because of the morphology of the coastline, and wave processes will be subdued by the protection afforded by the highly embayed shorelines. Process categories are thus predicted to range from tide dominated (T), to tide dominated, wave influenced (Tw), to tide dominated, wave influenced, fluvial affected (Twf).

Matrix 2 (Figure 9B) is a new addition to the Ainsworth et al. (2008) model and relates to basins with a high tidal resonance potential. That is, basins with relatively wide shelves (>75 km [>47 mi] wide) that have the potential to amplify the tidal wave via resonance effects (Figure 7). Note that in matrix 2 (Figure 9B), all except 10 of the classifications shown in the boxes are tide dominated. The ten exceptions occur in two boxes under high A/S conditions on straight to lobate shorelines. As previously discussed in the section on Influence of Accommodation and Sediment Supply, this is caused by the relatively high rates of accommodation development, potentially allowing wave and fluvial processes to dominate over the tidal processes. This is because under high rates of accommodation development, shelf water depths are relatively high, resulting in minimal wave frictional damping and also reduced tidal wave amplification.

An alternative method of displaying the predictive matrices shown in Figure 9 is via a decision tree (Figure 10). In the decision tree, queries relating to the tidal resonance potential of the basin, wave versus fluvial effectiveness, low versus high A/S ratios, and shoreline morphology can be posed, which leads to a prediction of the expected dominant and subordinate depositional processes that will occur at the shoreline.

Examples of the practical usage of these matrices and decision trees for predicting changes in coastal depositional processes through time and

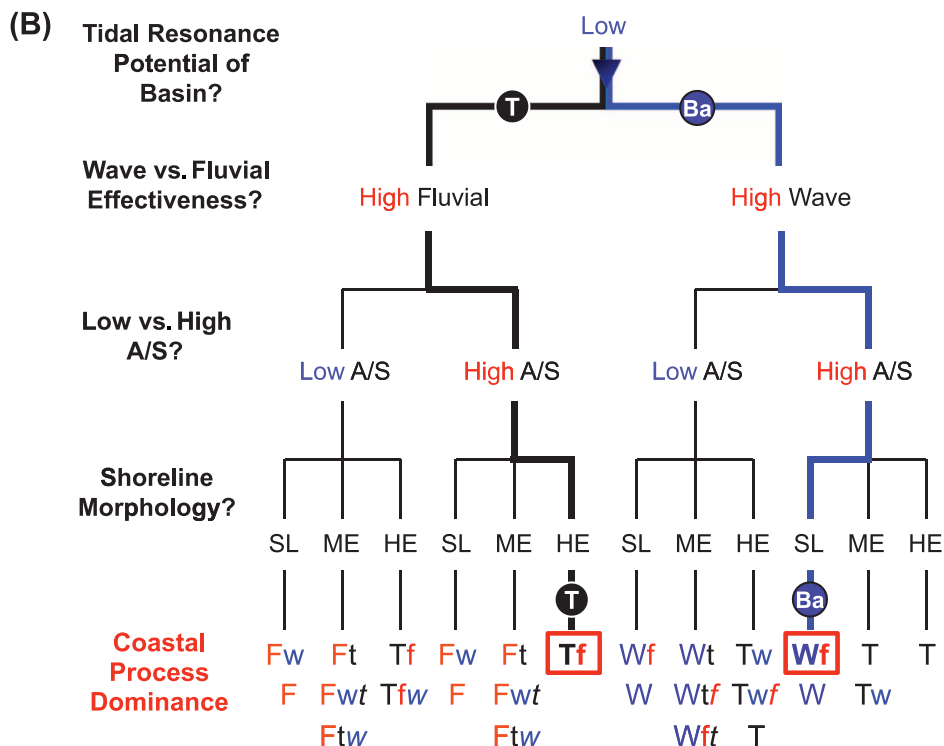
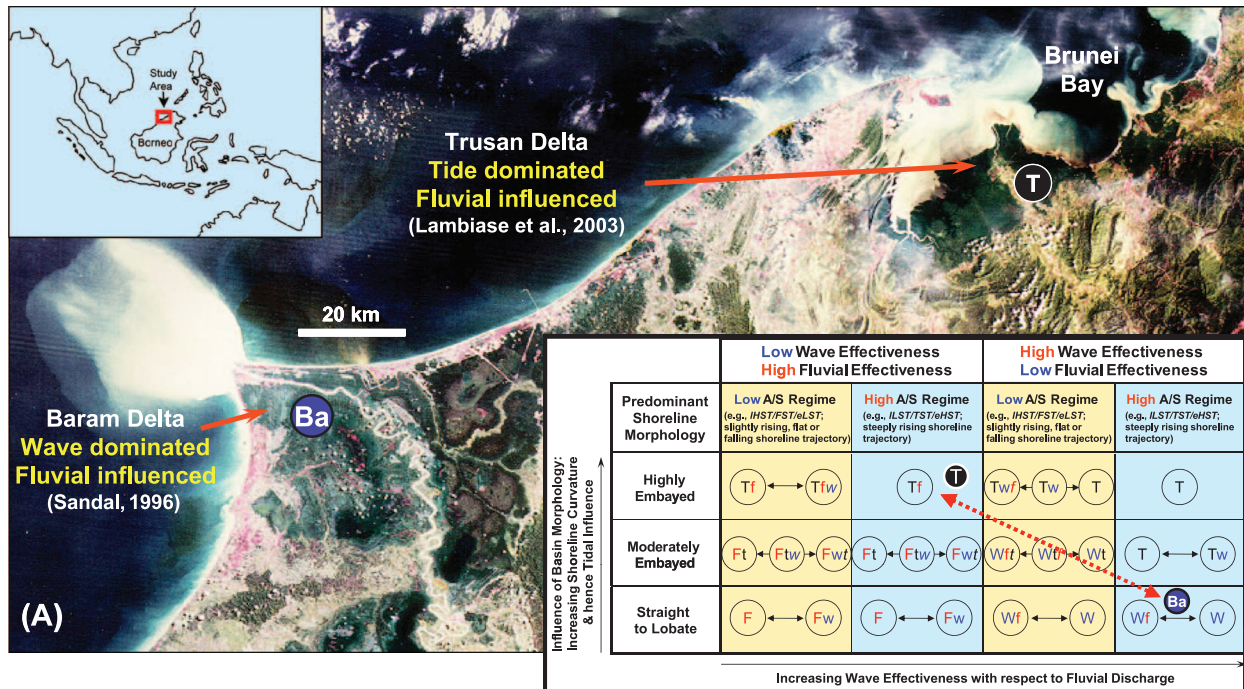


Figure 11. Spatial prediction of coastal process dominance. Using the low tidal resonance potential matrix (matrix 1; Figure 9), the Baram Delta (Ba), and the Trusan Delta (T) in panel A are predicted to be wave dominated, fluvial influenced (Wf), and tide dominated, fluvial influenced (Tf), respectively. The decision tree in panel B shows how by answering the queries posed, the process dominance at the shoreline can be predicted. See Figure 3 for definition of coastal process dominance acronyms. Inset matrix in panel A can be seen at a larger scale in Figure 9A.

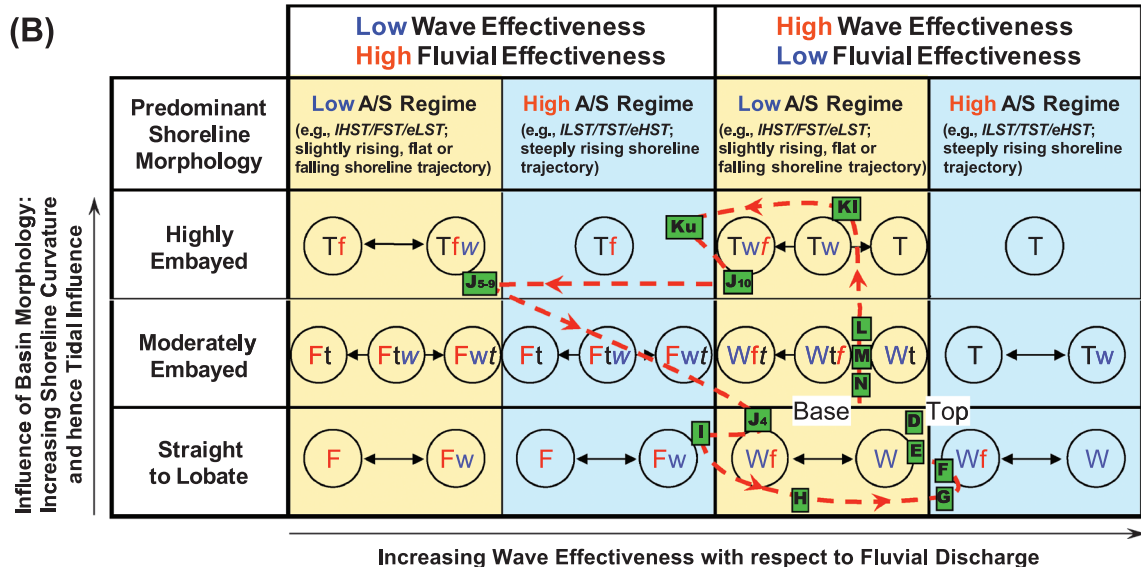
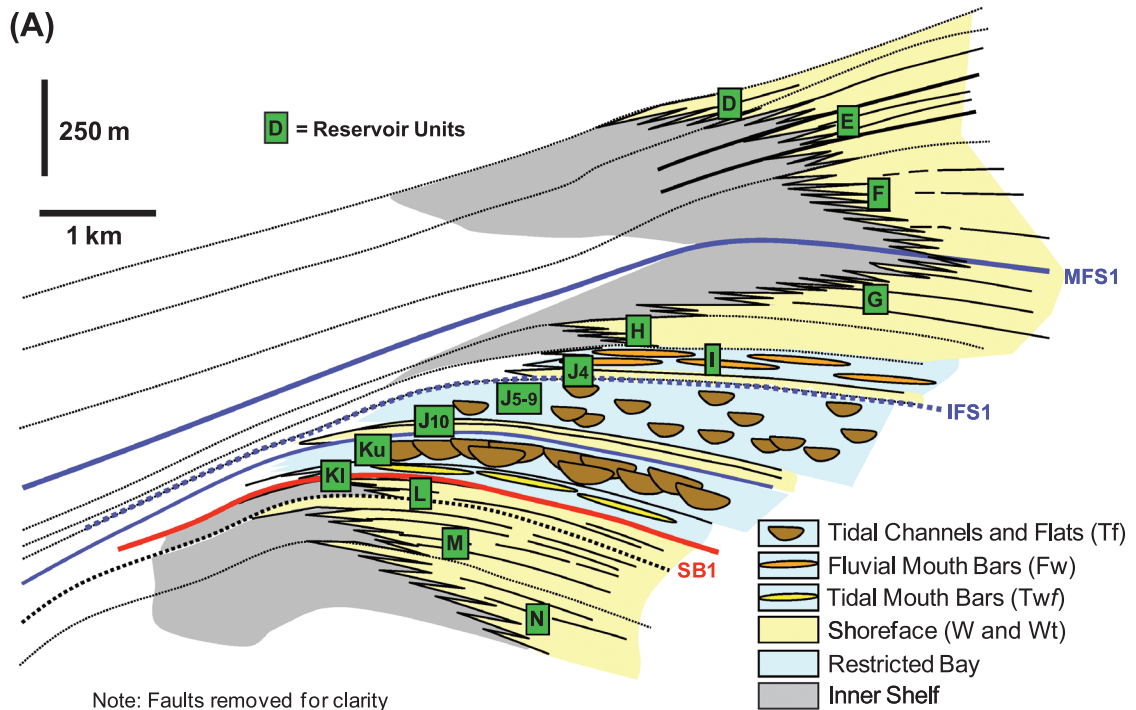


Figure 12. Temporal prediction of coastal process dominance. (A) Schematic cross section through the Champion field, Brunei, showing key sequence-stratigraphic surfaces, depositional environment interpretations, and reservoir units. Figure based on data presented by Hadley et al. (2006). (B) Low tidal resonance potential matrix (Matrix 1; Figure 9A) used to illustrate the evolution of the different reservoir units through time. Classification of the units in terms of depositional process was achieved through core interpretation. By plotting the units on the matrix, inferences regarding the evolution of the paleogeography of the area through time can also be made (see Predominant Shoreline Morphology on vertical axis). SB = sequence boundary; IFS = initial flooding surface; MFS = maximum flooding surface. See Figure 3 for definition of coastal process dominance acronyms.

Brunei Bay ([T] Figure 11). This geographic location suggests that a high likelihood exists that the system will be sheltered from direct oceanic wave approach and will therefore have a relatively low

wave effectiveness and a relatively high fluvial effectiveness. In terms of A/S regime, the same argument as used for the Baram Delta can be applied. This suggests that the system is most likely

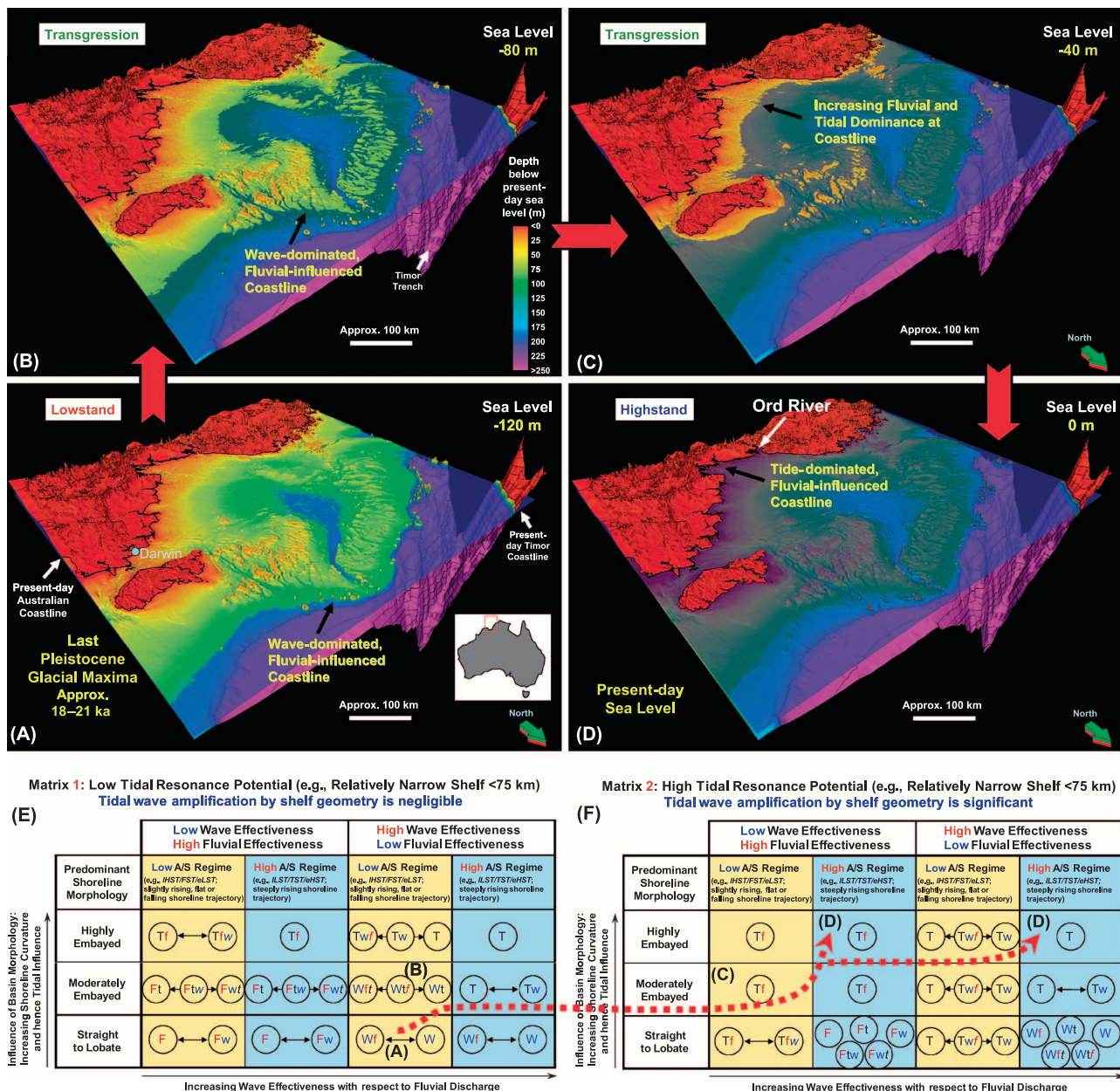


Figure 13. Spatial and temporal prediction of coastal process dominance. Three-dimensional rendering of bathymetry and topography data (courtesy of Geoscience Australia) from the Bonaparte Basin, northwestern Australia. The evolution of a transgression from the last glacial maxima, Pleistocene lowstand to the present-day Holocene highstand is shown in panels A to D. From a sea level low of -120 m (-394 ft) below present day (A), the shoreline can be tracked through time and space and can be seen to evolve from a straight to lobate, wave-dominated, fluvial-influenced (Wf), shelf margin delta (A) to a tide-dominated (T), and tide-dominated fluvial-influenced (Tf) present-day coastline (D). These changes can be predicted by using the matrices shown in panels E and F. See dotted red line. The shelf width passes the critical 75 km (47 mi) value between phases (B) and (C), and the matrix therefore switches from the narrow shelf matrix (E) to the wide shelf matrix (F). See Figure 3 for definition of coastal process dominance acronyms. Matrices in panels E and F are shown at full scale in Figure 9.

in a relatively high A/S setting. As can be seen from the satellite image (Figure 11A), Brunei Bay is a highly embayed-type setting. The prediction is therefore that the coastal depositional style would

be tide dominated fluvial influenced (Tf). Lambiasi et al. (2003) have studied the Trusan Delta and classify it as a tide dominated, fluvial influenced system.

Temporal Prediction

Figure 12 illustrates an example of how the matrices can be used to track changes in coastal process dominance through time. In this example from the Miocene deposits of the Champion field ([Brunei] Hadley et al., 2006), a series of cored marginal marine reservoir units have been classified according to the ternary plot in Figure 3C. The relatively low tidal resonance potential matrix (matrix 1; Figure 9A) is used because paleogeographic evidence suggests that a relatively narrow shelf existed at the time of deposition. The evolution of process changes at the shoreline can be plotted on the matrix using the process classification and the sequence-stratigraphic interpretation to define relative A/S regimes. Inferences can then be made regarding the shoreline morphology and the relative effectiveness of waves and fluvial systems. For example, at the base of the succession reservoirs, N to L are interpreted as wave dominated, tide influenced (Wt) and sit in a late highstand or low A/S regime (Figure 8). It can therefore be inferred that they were deposited in a moderately embayed setting that had high wave effectiveness. Reservoirs K1 to J5-9 are interpreted as tide-dominated systems (Twf and Tf, respectively) and fluctuate between low and high A/S regimes and low and high wave and fluvial influences. It can be inferred from Figure 12 that in all these reservoirs, the driving force behind the tidal influence was the highly embayed nature of the shoreline at the time of deposition (see predominant shoreline morphology on Figure 12B). With the exception of reservoir unit I (fluvial-dominated [Fw]), reservoirs J4 to D are interpreted as being wave dominated (W). They fluctuate between low and high A/S regimes. The fact that waves and rivers can dominate deposition suggests that the highly embayed shorelines of K1 to J5-9 have been straightened out (see predominant shoreline morphology on Figure 12B), perhaps by the shorelines prograding beyond the confines of the embayment. This would reduce the tidal influence on the systems and also bring the shorelines into direct exposure to oceanic waves.

Spatial and Temporal Prediction

Figure 13 illustrates an example of how the matrices can be used to predict both changes across space and through time. The example used here is from the Bonaparte Basin in northwestern Australia (Figure 13A) and shows the changes in coastal depositional process dominance experienced during the transgression from the last Pleistocene lowstand to the present-day Holocene highstand. Bathymetric data courtesy of Geoscience Australia are displayed in three dimensions, and paleo-sea levels are superimposed onto the topography at 120 m (75 mi) below present-day sea level (Figure 13A), -80 m below present-day sea level (Figure 13B), -40 m below present-day sea level (Figure 13C), and the present-day sea level (Figure 13D). Figure 13A represents the paleogeography at approximately the last glacial maxima (18–21 ka) and hence would represent a lowstand in sea level. The shelf can be seen to be virtually nonexistent at this time so the coastal systems would plot on matrix 1, the low tidal resonance potential matrix (Figures 9A, 13E). The shoreline can be seen to be straight to lobate, and no potential obstacle to oceanic waves impinging directly on the shoreline exists. The system would therefore be predicted to be wave dominated (W) or wave dominated, fluvial influenced (Wf). The geometry of the shelf-margin delta seen in Figure 13A is indeed suggestive of this mode of process domination. As the transgression progresses across the shelf, the effective width of the shelf increases to more than 75 km (>47 mi). This suggests that matrix 2 (Figures 9B, 13F), the high tidal resonance potential matrix, should come into play and that the coastal systems should become increasingly tide influenced and tide dominated. Panels A and B in Figure 13 illustrate the increasing width of the shelf area and the more rugose and highly embayed nature of the shoreline during the transgression. When plotted on the high tidal resonance potential matrix (Figure 13F), the final present-day sea level (Figure 13D) is predicted to be tide dominated (T) or tide dominated, fluvial influenced (Tf). The present-day coastline in the Bonaparte Basin is indeed macro-tidal and tide dominated; the well-studied Ord

River estuary with a spring tidal range of 5.9 m (Wright, 1985) is located in the Bonaparte Basin (Figure 13D).

In the three examples previously shown, the matrices can be used to visually and dynamically track changes in process dominance and subordination through time and across space while the decision trees are useful for logically breaking down the queries required to generate the process predictions.

EXPLORATION AND PRODUCTION IMPLICATIONS

The classification approach, predictive matrices, and decision trees have several practical exploration and production applications.

Uncertainty Management and Reduction

In the subsurface, a full suite of data is not commonly available so any method of prediction will have aspects of uncertainty (Smalley et al., 2008). The approach previously detailed provides a simple methodology for managing the subsurface uncertainty associated with predicting coastal depositional processes. Figure 14A shows one example of how the decision trees can be used to manage uncertainty. In this example, the geoscientist has information regarding the tidal resonance potential of the basin, the wave versus fluvial effectiveness and the A/S regime. However, no information is available as to the shoreline morphology at the time of deposition. This means that all three branches of the decision tree at this level must be selected to span the range of uncertainty presented by this absence of data. With the information available, the process regime at the shoreline could be (1) fluvial dominated, wave influenced (Fw); (2) fluvial dominated, tide influenced (Ft); (3) tide dominated, fluvial influenced (Tf); (4) fluvial dominated (F); (5) fluvial dominated, wave influenced, tide affected (Fwt); or (6) fluvial dominated, tide influenced, wave affected (Ftw).

Once the range of potential process domination and subordination regimes has been pre-

dicted, depositional environment, facies, sand/shale ratio, geometric and other data can then be collated for each of these process domination regimes (Figure 2). This information can then be used to generate exploration or production type scenarios. For example, in an exploration setting, collated data from the specified process settings could be used to provide ranges in sand/shale or net/gross ratios for volumetric estimates. Or collated geometric data could be used to apply reservoir continuity predictions that can be related to stratigraphic connectivity (Ainsworth, 2005; 2010). In a production- or development-type setting, the geometric data from the six potential process domination regimes could be used to generate very different but realistic depositional scenarios for three-dimensional (3-D) reservoir models.

If no further information is available, then the six potential scenarios in Figure 14A would be considered equiprobable. However, sometimes, subsurface uncertainty exists, but some information indicates a higher probability of occurrence of one potential scenario over another (Smalley et al., 2008). In this case, probabilities can be assigned to branches of the decision tree, and probabilities can be multiplied through to produce an end probability of each potential process dominance regime occurring. These probabilities can then be further used in exploration risking or weighting of 3-D reservoir modeling scenarios.

If, after considering all the available data, the range of existing uncertainty in, for example, net/gross ratios or geometries of sand bodies under the different process regimes is too great to make the required decision, then the decision trees can also be used to highlight parameters for which more data need to be acquired to reduce uncertainty to acceptable levels. For example, if in Figure 14A the range of geometries of sand bodies or net/gross ratios deposited under Fw, Ft, Tf, F, Fwt, or Ftw process-domination conditions was too great to allow a project to progress, then this diagram would indicate that more information on the shoreline morphology would potentially reduce this uncertainty in process domination. A case could therefore be made to further appraise a prospect or field by potentially acquiring data to reduce uncertainty

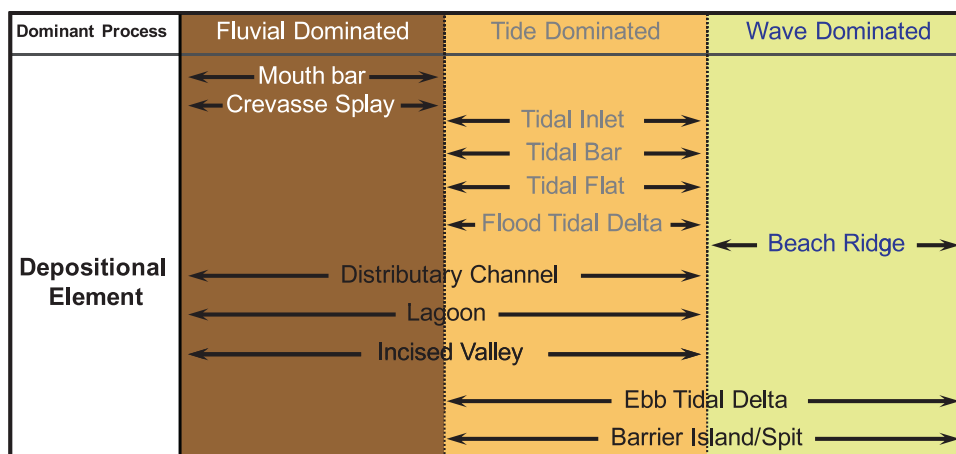


Figure 15. Process domination and depositional elements. This figure shows a number of potential depositional elements and environments from marginal marine systems. It demonstrates how individual depositional elements or environments can potentially occur under more than one process domination regime. For example, distributary channels can be classified as either fluvial dominated or tide dominated. The importance of this distinction is that distributary channels may have very different geometries if they are fluvial or tide dominated. This highlights the importance of steps 2 and 3 in Figure 2 where once the dominant depositional processes have been identified (the subject of this article), the next step is to identify the environment of deposition and then to delineate geometries and/or properties for that environment. This is the subject of ongoing research (Vakarelov et al., 2010).

in paleogeographic setting. For example, acquisition of a high-resolution 3-D seismic survey, more information on biostratigraphy, or acquiring core to reduce uncertainty in process domination may be suggested. The value of information analyses could be run on all these potential options.

A key advantage of the approach detailed in this article is that it provides a fully consistent and auditable method for defining process dominance along a shoreline. Decisions made regarding tidal resonance potential of a basin, wave versus fluvial effectiveness, A/S regime, and shoreline morphology can be mapped on the decision trees, and a record is kept as to why predictions or appraisal decisions were made and why scenarios were chosen.

In addition, the matrices (Figure 9) can be used as visualization tools to plot the evolution and changes in coastal depositional process dominance within or between reservoirs through time and/or across space (Figures 12, 13).

Improved Regional Prediction

The decision trees can be used from top down or bottom up. If process regime at a shoreline is being predicted, then the decision trees are used

in a top-down approach as per the previous examples discussed. However, if the process regime is already known, for example, a core has been interpreted, the decision trees can be used in reverse. Figure 14B provides a good example of how reverse engineering can be applied to determine more information about regional setting. In this figure, the example of the Mitchell River Delta from the Gulf of Carpentaria (Australia; Figure 6) is used. As previously discussed, the Mitchell River Delta is interpreted as a tide-dominated, fluvial-influenced, wave-affected system ([Tfw] Figure 6). Figure 14B shows how this process regime can only occur under two combinations of parameters (blue boxes in Figure 14B). Hence, the other regional parameters can be predicted; shoreline morphology, A/S regime, wave versus fluvial effectiveness, and tidal resonance potential of the basin (thick black line on the left-hand and thick blue line on Figure 14B). Both predictions suggest low A/S regimes and high fluvial effectiveness. However, tidal resonance potential and shoreline morphology are different. Two different potential combinations of parameters can therefore be seen to potentially generate the same process dominance regime at the shoreline.

In the case of the Mitchell River Delta (Figure 6), the actual parameters are known and plotted as the thick blue line on Figure 14B. The shoreline morphology can be seen to be straight to lobate (Figure 6); the A/S regime is known to be low with a very shallow shelf in front of the Mitchell River Delta with water depths reaching only 20 m (66 ft) approximately 30 km (~19 mi) from the coastline; the high fluvial effectiveness is demonstrated by the large distributary channel and mouth bar (Figure 6); the inset map in Figure 6 also demonstrates the very wide shelf area of the Gulf of Carpentaria, which means that this basin has a high tidal resonance potential. The method of predicting more regional parameters can therefore be seen to work in this modern example, which suggests that it will also have value for predicting regional parameters in the ancient if process controls can be interpreted from cores or outcrops.

The Champion field shown in Figure 12 is also a good example of how the matrices can be used to determine information regarding paleogeography. In this example, process domination regimes were determined from core. This information along with A/S data determined from sequence-stratigraphic interpretations enabled the individual reservoir units to be plotted on the low tidal resonance potential matrix (Figure 9A) and inferences to be made about evolving paleogeography through time.

Using the decision trees also permits self-consistency checks to be made. For example, if a core were interpreted to be tide dominated, fluvial influenced ([Tf] Figure 3) and the tidal resonance potential of the basin was interpreted to be low, but the shoreline was interpreted to be straight to lobate, then this combination of possibilities would not be consistent with the decision tree (Figure 14B; thick black lines). One would then be forced to check one's assumptions regarding the morphology of the shoreline because the tree indicates that both occurrences of Tf are associated with highly embayed shorelines. By the same degree, if uncertainty occurs pertaining to the formative processes of the sedimentary structures in the core (see section on Application of Classification to Ancient Systems), several possible coastal process dominance combinations could be selected and

the self-consistency cross checked with the other known data in the basin by moving vertically upward through the decision tree.

FURTHER WORK

Depositional Environments and Elements

As mentioned in the introduction, this article has focused on predicting process dominance at the shoreline, which is the first step in analyzing and predicting the reservoir architecture of coastal systems (Figure 2). The second step is then predicting potential depositional environments and geometries of depositional elements (sand bodies or heterogeneities) that occur under the process conditions imposed upon the shorelines (Figure 2). Figure 15 provides an example of some potential depositional elements that may be found in marginal marine settings. The figure illustrates how these individual elements may also occur under more than one process domination regime. For example, distributary channels may be classified as being fluvial dominated or in some cases tide dominated. This is important because the geometries of the same element (distributary channels in this case) under different process conditions are likely to be very different. The process classification and prediction workflow demonstrated herein provides a framework for the population of the geometric data that can be collated for the different potential reservoir elements (Vakarelov et al., 2010).

The matrices in Figure 9 illustrate graphically how the same end-process domination conditions can occur under different A/S regimes. For example, in matrix 1 (Figure 9A), the two columns on the left (low A/S and high A/S) show almost identical process domination products in both low and high A/S regimes, with the only differences being when the shorelines are highly embayed. In these circumstances, it is still necessary to consider the low and high A/S settings separately as the same reservoir elements under different A/S regimes can produce markedly different depositional geometries and architectures. Enge and Howell (2010) demonstrated how the dip angle on fluvial-dominated

mouth-bar clinoforms varied significantly from low to high A/S settings. In reservoir simulation studies of outcrops from these two A/S settings, a significant difference in sand-body connectivity and hence reservoir sweep efficiency was observed.

Further work is also required on the application of process dominance analyses to mudstones. In the modern, processes that dominate in areas of muddy substrates can be readily observed, measured, and classified. However, in the ancient, mudstones have historically been assumed to have been deposited from fallout of suspension. Only recently have mudstones been analyzed in terms of the more complex range of processes that govern their deposition (Nittrouer and Wright, 1994). Because mudstones and siltstones form an important proportion of many coastal depositional systems, good analysis techniques and methodologies are required to attribute dominant and subordinate depositional processes to mudstones. Close attention to ichnological techniques in these types of settings is also critical for determining depositional processes (MacEachern and Bann, 2008).

CONCLUSIONS

The key conclusions derived from this study are as follows:

1. A process-based classification of clastic coastal systems based on relative dominance of fluvial, tidal, and wave processes is proposed. The scheme permits semiquantitative classification of primary, secondary, and tertiary processes acting on shoreline depositional systems. The classification scheme provides a pragmatic coherent framework for the development of further studies of clastic coastal depositional systems and enables the direct comparison of both simple and complex mixed-process shoreline deposits.
2. Two matrices, one for low tidal resonance potential and one for high tidal resonance potential, can be used to predict changes in coastal process dominance through time and space.
3. These matrices can also be represented as a decision tree, which can be used to provide an audit trail for process predictions.

4. The decision tree can also be used to manage subsurface uncertainty when certain pieces of key information are unknown. As such, it can be used to define critical information required to reduce uncertainty and hence refine appraisal and development drilling strategies.
5. The matrices and decision tree can also be used in reverse. For example, if depositional processes are known from core or outcrop information, reverse engineering can produce predictions of shoreline morphology, A/S, wave versus fluvial effectiveness, and tidal resonance potential of a basin.
6. Combinations of processes acting at shorelines are well understood to determine reservoir architecture. The classification presented herein provides a framework for the further research that is required to understand how these process-dominance regimes convert into geometries, distributions, and properties of reservoir sand bodies and reservoir heterogeneities.

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